

# Algebraic Geometry – Exercise Sheet 3

Week 3: Schemes, morphisms, gluing and first global properties  
Monday and Thursday afternoon sessions; Lectures 11–15

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**Difficulty.** ★ routine; ★★ synthesis; ★★★ challenge or extension.

**Use.** The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

## Monday afternoon — material available: Lectures 1–11

### Exercise 1. Global functions as morphisms to the affine line

★ **CORE**

Let  $X$  be a scheme.

- (i) Choose an affine open covering  $X = \bigcup_i U_i$ . Explain why a global section  $s \in \Gamma(\mathcal{O}_X)$  determines compatible morphisms  $U_i \rightarrow \mathbb{A}_{\mathbb{Z}}^1$  and hence a morphism

$$f_s : X \longrightarrow \mathbb{A}_{\mathbb{Z}}^1.$$

- (ii) Show that

$$\mathrm{Hom}(X, \mathbb{A}_{\mathbb{Z}}^1) \simeq \Gamma(\mathcal{O}_X).$$

- (iii) If  $X = \mathrm{Spec}(A)$ , describe the morphism associated with  $a \in A$ .

### Exercise 2. The reduced point and the double point

★ **CORE**

Inside  $\mathbb{A}_{\mathbb{k}}^1 = \mathrm{Spec}(\mathbb{k}[t])$ , consider

$$P = \mathrm{Spec}(\mathbb{k}[t]/(t)), \quad D = \mathrm{Spec}(\mathbb{k}[t]/(t^2)).$$

- (i) Show that the two closed immersions have the same image as subsets of  $\mathbb{A}_{\mathbb{k}}^1$ .  
(ii) Show that  $P$  and  $D$  are not isomorphic as schemes.  
(iii) Compute the maps on local rings at their unique points.

### Exercise 3. A morphism between affine schemes

★ **CORE**

Let

$$\varphi : \mathbb{k}[u, v] \longrightarrow \mathbb{k}[t], \quad u \longmapsto t, \quad v \longmapsto t^2.$$

- (i) Describe the induced morphism  $f : \mathbb{A}_{\mathbb{k}}^1 \rightarrow \mathbb{A}_{\mathbb{k}}^2$ .  
(ii) Determine  $\ker(\varphi)$  and factor  $f$  through a closed subscheme of  $\mathbb{A}_{\mathbb{k}}^2$ .  
(iii) Show that the resulting morphism onto that closed subscheme is an isomorphism.

### Exercise 4. Gluing ideal sheaves and closed subschemes

★★ **CORE**

Let  $X = \bigcup_i U_i$  be an open covering. Suppose that each  $U_i$  carries a quasi-coherent ideal sheaf  $\mathcal{I}_i \subset \mathcal{O}_{U_i}$  and that

$$\mathcal{I}_i|_{U_i \cap U_j} = \mathcal{I}_j|_{U_i \cap U_j}$$

inside  $\mathcal{O}_X|_{U_i \cap U_j}$ .

- (i) Show that the  $\mathcal{I}_i$  glue to an ideal sheaf  $\mathcal{I} \subset \mathcal{O}_X$ .  
(ii) Prove that  $\mathcal{I}$  is quasi-coherent.  
(iii) If  $Z_i \hookrightarrow U_i$  is the closed subscheme defined by  $\mathcal{I}_i$ , show that the  $Z_i$  glue to the closed subscheme of  $X$  defined by  $\mathcal{I}$ .  
(iv) Explain why none of these assertions requires  $X$  to be affine.

### Exercise 5. Open subschemes and the punctured plane

★★ **CORE**

Let  $X$  be a scheme and let  $U \subset X$  be open.

- (i) Prove directly from the locally affine definition that  $(U, \mathcal{O}_X|_U)$  is a scheme.

(ii) Show that for every  $x \in U$  one has

$$\mathcal{O}_{U,x} \simeq \mathcal{O}_{X,x}, \quad \kappa_U(x) \simeq \kappa_X(x).$$

(iii) Cover the punctured affine plane  $P = \mathbb{A}_{\mathbb{k}}^2 \setminus \mathbb{W}(x, y)$  by two affine open subschemes.

(iv) Compute  $\Gamma(\mathcal{O}_P)$  and prove that  $P$  is not affine.

## Thursday afternoon — material available: Lectures 1–14

### Exercise 6. Dual numbers and tangent vectors

★★ CORE

Let  $D = \text{Spec}(\mathbb{k}[\varepsilon]/(\varepsilon^2))$  and let  $p = (a, b) \in \mathbb{A}_{\mathbb{k}}^2(\mathbb{k})$ .

(i) Show that morphisms  $D \rightarrow \mathbb{A}_{\mathbb{k}}^2$  whose closed point maps to  $p$  are uniquely of the form

$$x \mapsto a + u\varepsilon, \quad y \mapsto b + v\varepsilon.$$

(ii) Let  $C = \mathbb{W}(f) \subset \mathbb{A}_{\mathbb{k}}^2$ . Show that such a morphism factors through  $C$  precisely when

$$f(a, b) = 0, \quad f_x(a, b)u + f_y(a, b)v = 0.$$

(iii) Apply this to the parabola  $y = x^2$ .

(iv) Assume  $\text{char}(\mathbb{k}) \neq 2, 3$ . For the cusp  $C = \mathbb{W}(y^2 - x^3)$ , compare the tangent spaces at  $(0, 0)$  and  $(1, 1)$ . Which point is singular and which is smooth according to the Jacobian criterion?

### Exercise 7. The line with two origins

★★ CORE

Glue two copies

$$X_1 = \text{Spec}(\mathbb{k}[t_1]), \quad X_2 = \text{Spec}(\mathbb{k}[t_2])$$

along  $D(t_1) \simeq D(t_2)$  by  $t_1 \mapsto t_2$ . Let  $X$  be the resulting scheme.

(i) Describe the underlying topological space.

(ii) Show that  $\Gamma(\mathcal{O}_X) \simeq \mathbb{k}[t]$ .

(iii) Show that the canonical map  $X \rightarrow \text{Spec } \Gamma(\mathcal{O}_X)$  identifies the two origins.

(iv) Deduce that  $X$  is not affine.

### Exercise 8. Rational and closed points of the projective line

★★ CORE

Construct  $\mathbb{P}_{\mathbb{k}}^1$  by gluing  $U_0 = \text{Spec}(\mathbb{k}[t])$  and  $U_1 = \text{Spec}(\mathbb{k}[s])$  along  $s = t^{-1}$ .

(i) For every field extension  $K/\mathbb{k}$ , prove that

$$\mathbb{P}_{\mathbb{k}}^1(K) = \mathbb{P}_K^1.$$

Describe the points in the two standard charts.

(ii) Describe all closed points of  $\mathbb{P}_{\mathbb{k}}^1$ , compute their residue fields, and determine which are  $\mathbb{k}$ -rational.

(iii) Prove that  $\Gamma(\mathcal{O}_{\mathbb{P}_{\mathbb{k}}^1}) = \mathbb{k}$ .

(iv) Deduce that  $\mathbb{P}_{\mathbb{k}}^1$  is not affine.

### Exercise 9. The projective conic from its standard opens

★★ CORE

Let

$$S = \mathbb{k}[x_0, x_1, x_2]/(x_0x_2 - x_1^2), \quad C = \text{Proj}(S).$$

(i) Compute  $(S[x_0^{-1}])_0$  and  $(S[x_2^{-1}])_0$ .

(ii) Identify  $D_+(x_0)$  and  $D_+(x_2)$  with affine lines.

(iii) Describe their overlap and recover the gluing rule for  $\mathbb{P}^1$ .

(iv) Show that projection from  $p = [1 : 0 : 0]$  extends to a morphism  $\pi : C \rightarrow \mathbb{P}_{\mathbb{k}}^1$  by using the formulas  $[x_0 : x_1]$  on  $D_+(x_0)$  and  $[x_1 : x_2]$  on  $D_+(x_2)$ . Prove that its inverse is  $[s : t] \mapsto [s^2 : st : t^2]$ .

**Exercise 10. Base loci: projection and Cremona**

\*\*\* CHOICE

- (i) Consider

$$\pi : \mathbb{P}_{\mathbb{k}}^2 \dashrightarrow \mathbb{P}_{\mathbb{k}}^1, \quad [x_0 : x_1 : x_2] \longmapsto [x_1 : x_2].$$

Determine its domain of definition and its base locus.

- (ii) In  $\mathbb{P}_{\mathbb{k}}^2 \times \mathbb{P}_{\mathbb{k}}^1$ , show that the closure of its graph is defined by  $x_1v - x_2u = 0$ . Describe the fibre of the first projection over  $[1 : 0 : 0]$ .
- (iii) For the standard Cremona transformation

$$[x_0 : x_1 : x_2] \dashrightarrow [x_1x_2 : x_0x_2 : x_0x_1],$$

compute the base locus and show that the transformation is its own inverse on the dense torus  $x_0x_1x_2 \neq 0$ .**Exercise 11. Connectedness beyond affine schemes**

\*\* CORE

Let  $X$  be a scheme.

- (i) Prove that  $X$  is disconnected if and only if  $\Gamma(\mathcal{O}_X)$  has a non-trivial idempotent.
- (ii) Use the global-function computations from this sheet to show that  $\mathbb{P}_{\mathbb{k}}^1$  and the line with two origins are connected.
- (iii) Compare the connected reducible scheme  $\text{Spec}(\mathbb{k}[x, y]/(xy))$  with the disconnected scheme  $\text{Spec}(\mathbb{k} \times \mathbb{k})$ .

**Exercise 12. A doubled projective line and its reduction**

\*\*\* CORE

Let

$$U_0 = \text{Spec}(\mathbb{k}[t, \varepsilon]/(\varepsilon^2)), \quad U_1 = \text{Spec}(\mathbb{k}[s, \eta]/(\eta^2)).$$

Glue  $D(t) \subset U_0$  to  $D(s) \subset U_1$  using  $s = t^{-1}$  and  $\eta = \varepsilon$ , and call the resulting scheme  $X$ .

- (i) Show that  $(\varepsilon) \subset \mathcal{O}_{U_0}$  and  $(\eta) \subset \mathcal{O}_{U_1}$  glue to a quasi-coherent ideal sheaf  $\mathcal{I} \subset \mathcal{O}_X$  with  $\mathcal{I}^2 = 0$ .
- (ii) Describe  $X_{\text{red}}$  and prove that  $X_{\text{red}} \simeq \mathbb{P}_{\mathbb{k}}^1$ .
- (iii) Show that every morphism from a reduced scheme  $Y$  to  $X$  factors uniquely through  $X_{\text{red}} \hookrightarrow X$ .
- (iv) Deduce that  $X$  is not affine.

**Exercise 13. A weighted projective chart**

\*\*\* OPTIONAL

Let  $R = \mathbb{k}[x, y, z]$  with  $\deg x = \deg y = 1$  and  $\deg z = 2$ , and set  $X = \text{Proj}(R)$ . Show that  $D_+(x)$  and  $D_+(y)$  are affine planes and that

$$D_+(z) \simeq \text{Spec}(\mathbb{k}[a, b, c]/(ac - b^2)).$$

Identify the singular point of this chart.

**After Friday's lecture or preparation for Week 4****Exercise 14. Components and generic points**

\*\* CHOICE

Let

$$I = (xy, xz) \subset \mathbb{k}[x, y, z], \quad X = \text{Spec}(\mathbb{k}[x, y, z]/I).$$

Show that  $I = (x) \cap (y, z)$ . Determine the irreducible components, their generic points and dimensions, and decide whether  $X$  is connected or integral.**Exercise 15. The function field of the cusp**

\*\* CHOICE

Let

$$C = \text{Spec}(\mathbb{k}[x, y]/(y^2 - x^3)).$$

Show that  $C$  is integral and that  $K(C) \simeq \mathbb{k}(t)$ .